Questions

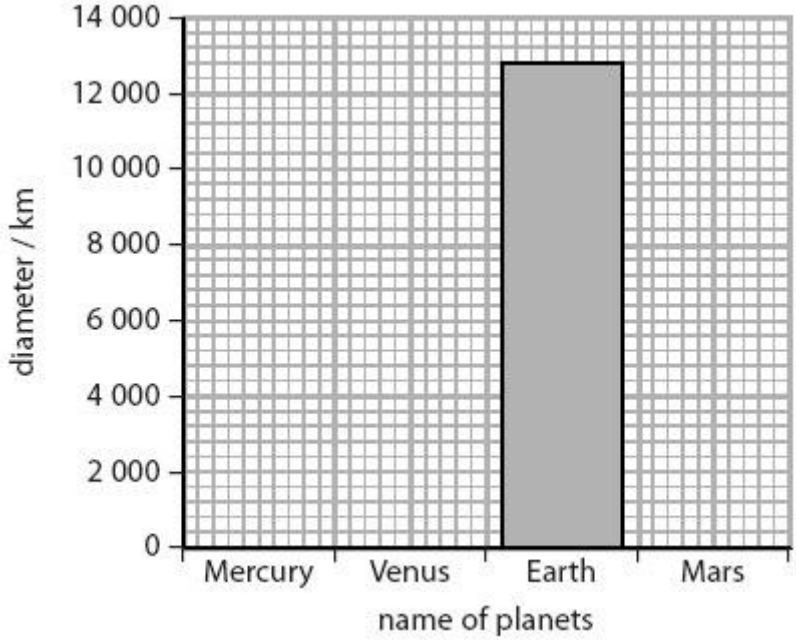
Q1.

The table gives information about the diameters and distances of the four planets closest to the Sun.

planet	distance from the Sun / AU	diameter of the planet / km
Mercury	0.39	4 900
Venus	0.72	12 100
Earth	1.00	12 800
Mars	1.52	6 800

(i) Put the information about the diameter of the planets on to the bar chart.
The diameter for Earth has been done for you.

(2)



(ii) The distance of the planets from the Sun has been given in Astronomical Units (AU).
1 AU is 150 000 000 km.
Calculate the distance of Mars from the Sun in kilometres.

(2)

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Q2.

The table gives information about the diameters and distances of the four planets closest to the Sun.

(i) Put the information about the diameter of the planets on to the bar chart.

The diameter for Earth has been done for you.

(2)

(ii) The distance of the planets from the Sun has been given in Astronomical Units (AU).
1 AU is 150 000 000 km.
Calculate the distance of Mars from the Sun in kilometres.

(2)

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Q3.

* Describe how modern telescopes have contributed to our understanding of the Universe.

(6)

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Q4.

The Sun is at the centre of our Solar System.

(i) Complete the following sentence.

(1)

Our Solar System is near the edge of a galaxy called the .....

(ii) Complete the sentence by putting a cross (☒) in the box next to your answer.

When the Sun nears the end of its life it will become a

(1)

☐ **A** black hole

☐ **B** neutron star

☐ **C** supernova

☐ **D** white dwarf

Q5.

(i) A long time ago, scientists believed that the Earth was at the centre of the Solar System.

Evidence has since proved that the Sun is at the centre of the Solar System.
State **one other** idea about the Solar System that **has** changed over time.

(1)

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(ii) Figure 13 shows data for some of the planets of the Solar System.

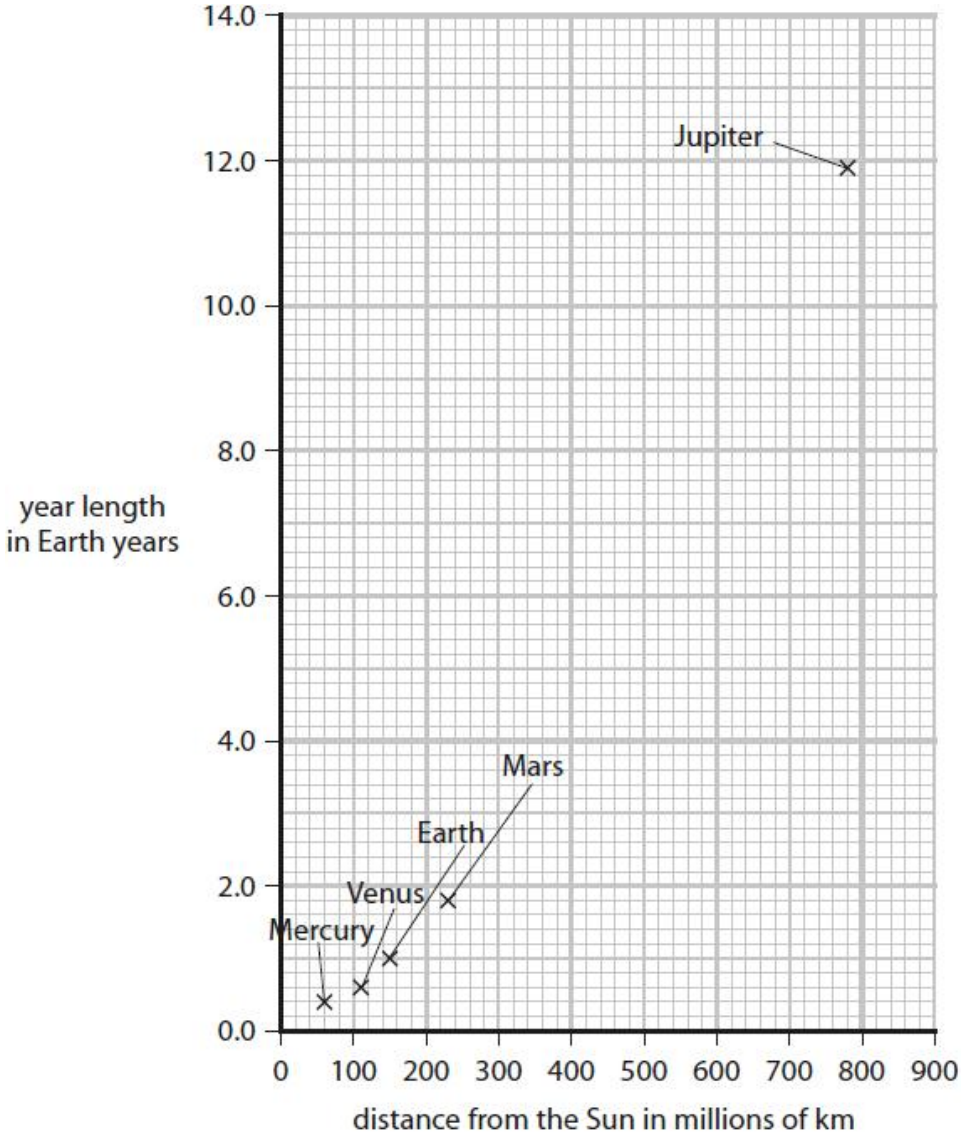

Figure 13

Ceres is an asteroid that orbits the Sun between Mars and Jupiter. It takes Ceres 4.6 Earth years to make one orbit of the Sun.

Use the graph to estimate the distance of Ceres from the Sun.

Show your working.

(3)

distance of Ceres from the Sun = ... millions of km

(Total for question = 4 marks)

Q6.

Satellites are used to gather data about the origin of the Universe.

The Big Bang theory is a theory about the origin of the Universe.

Evidence for the Big Bang theory is provided by red-shift and CMB radiation.

(i) Describe what is meant by red-shift.

(2)

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(ii) Explain how red-shift provides evidence for the Big Bang theory.

(2)

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(iii) The Cosmic Background Explorer (COBE) satellite observed CMB radiation from 1989 to 1993.
State what the 'M' in CMB radiation stands for.

(1)

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(iv) State what is meant by 'cosmic background radiation'.

(1)

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(v) Explain how the presence of CMB radiation provides evidence for the Big Bang theory.

(2)

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(Total for question = 8 marks)

Q7.

During the twentieth century red-shift and CMB radiation were discovered.
They have provided scientists with data to test theories of the origin of the Universe.

(i) Complete the following sentence.

(1)

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(ii) State which theory about the origin of the Universe is supported by the existence of CMB.

(1)

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(iii) There is a red-shift in the light received from some galaxies.
State what is meant by red-shift.

(1)

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(iv) Some galaxies show greater red-shift than others.
Explain what this suggests about the Universe.

(2)

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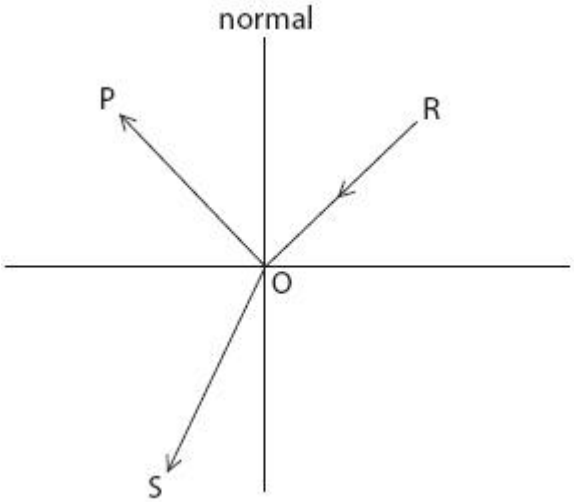
Q8.

When light strikes a glass surface it can be both refracted and reflected.
(a) The diagram shows the possible paths for a ray of light which strikes a surface at the point O.



(i) Which of the lines show the possible path of a ray of light passing from air into glass?
Put a cross ( ) in the box next to your answer.

(1)

☐ **A** POS

☐ **B** POR

☐ **C** ROP

☐ **D** ROS

(ii) The diagram shows a water wave going from deep water into an area of much shallower water. The wave is refracted at the boundary between deep water and shallow water.

Which row of the table is correct for what happens when the wave is refracted?
Put a cross ( ) in the box next to your answer.

(1)

	speed	direction
☐ A	stays the same	changes
☐ B	stays the same	stays the same
☐ C	changes	changes
☐ D	changes	stays the same

(b) In 1610 Galileo used a refracting telescope to observe the planet Jupiter..

(i) Explain how a refracting telescope produces a magnified image of Jupiter.

(3)

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(ii) In 1610, the geocentric model of the Solar System was commonly accepted.

Explain how Galileo's observations contradicted the geocentric model.

(3)

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(c) Light travels the 150 million km from the Sun to the Earth in about 500 s.
It takes about 2100 s for light to reach the Earth from Jupiter.
Using this information, calculate the approximate distance of Jupiter from the Earth.

(2)

distance of Jupiter from the Earth =.....million km

(Total for Question = 10 marks)

Q9.

(a) The Sun is at the centre of our Solar System.

(i) Complete the following sentence.

(1)

Our Solar System is near the edge of a galaxy called the .....

(ii) Complete the sentence by putting a cross (☒) in the box next to your answer.

When the Sun nears the end of its life it will become a

(1)

☐ **A** black hole

☐ **B** neutron star

☐ **C** supernova

☐ **D** white dwarf

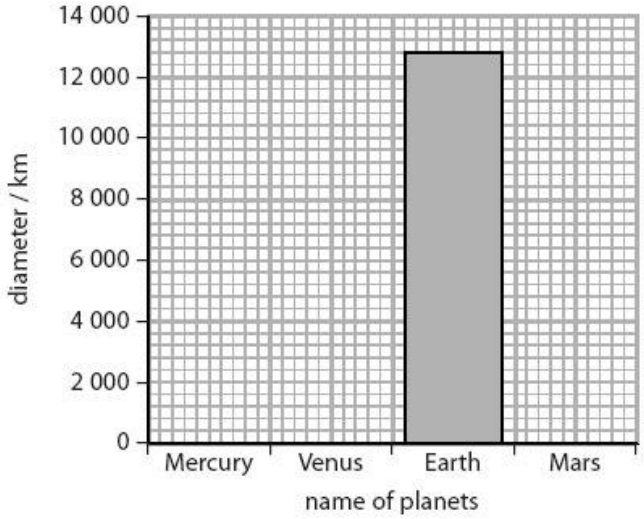
(b) The table gives information about the diameters and distances of the four planets closest to the Sun.

planet	distance from the Sun / AU	diameter of the planet / km
Mercury	0.39	4 900
Venus	0.72	12 100
Earth	1.00	12 800
Mars	1.52	6 800

(i) Put the information about the diameter of the planets on to the bar chart.

The diameter for Earth has been done for you.

(2)



(ii) The distance of the planets from the Sun has been given in Astronomical Units (AU).
1 AU is 150 000 000 km.
Calculate the distance of Mars from the Sun in kilometres.

(2)

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*(c) For many years scientists have searched for evidence of intelligent life in our Solar System and in the rest of the Universe.

Describe the methods scientists have used to help with this search in both our Solar System and the rest of the Universe.

(6)

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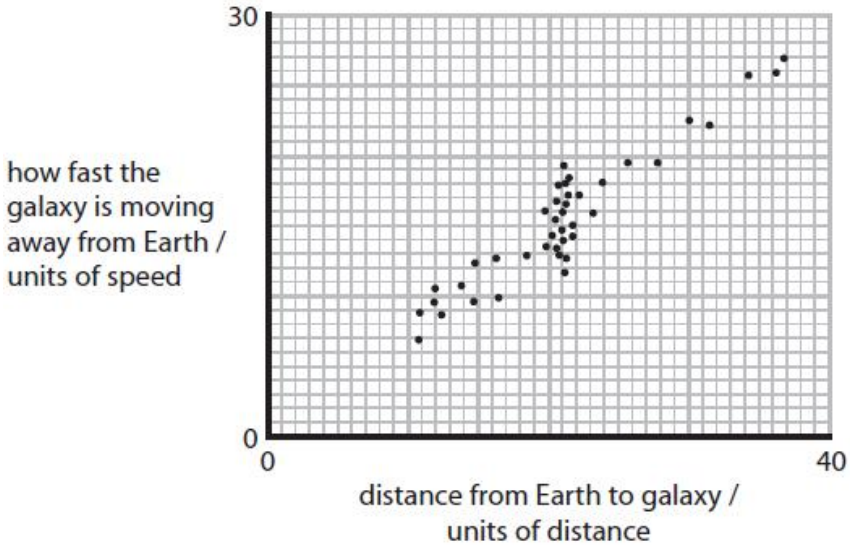
(Total for Question = 12 marks)

Q10.

(a) Hubble measured the distance of many galaxies from Earth.
He also measured the speed at which each galaxy moved away from Earth.

Hubble plotted his data on a graph like this.



(i) Plot the point: distance = 5 units, speed = 4 units

(1)

(ii) Draw the straight line of best fit.

(1)

(b) Hubble's work led to the theory of the Big Bang.

Describe what is meant by the Big Bang theory.

(2)

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Q11.

(a) Our Sun is a star.

(i) The life cycle of a star has several stages.
The Sun is in its main sequence stage.
Complete the sentence by putting a cross (☒) in the box next to your answer.
The next stage in the life cycle of the Sun will be

(1)

☐ **A** a nebula
☐ **B** a red giant
☐ **C** a white dwarf
☐ **D** a black hole

(ii) Complete the sentence by putting a cross (☒) in the box next to your answer.
The galaxy which contains our Sun is called

(1)

☐ **A** Andromeda
☐ **B** Orion
☐ **C** the Milky Way
☐ **D** the Solar System

(iii) Which of these is furthest from the Sun?
Put a cross (☒) in the box next to your answer.

(1)

- ☐ **A** Earth
- ☐ **B** The Moon
- ☐ **C** The planet Jupiter
- ☐ **D** The star Proxima Centauri

(b) Scientists search for signs of life on the planet Mars.
Describe **one** method they use to search for signs of life on the planet Mars.

(2)

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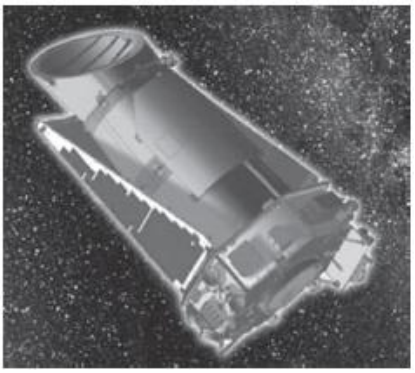
(c) Scientists search for new planets orbiting distant stars.
The photograph shows the Kepler space telescope.



The Kepler space telescope records the brightness of light from distant stars.
When a planet passes in front of a distant star, there is a tiny dip in the brightness of the light from the star.

(i) Many telescopes are used on the Earth but the Kepler space telescope orbits above the Earth's atmosphere.
Explain why telescopes that search for planets are not on the Earth's surface.

(2)

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(ii) The Kepler space telescope was pointed towards one star.
It detected the same dip in the brightness of the star every 150 days.
Suggest what information this gives about the planet that orbits this star.

(1)

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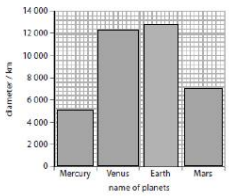
(Total for Question = 8 marks)

Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
(i)	 Mercury 4 900 km Venus 12 100 km Mars 6 800 km Three correct (2) One or two correct (1) + or – one square Judge by eye		**(2)**
(ii)	Correct information from table 1.52 (1) Conversion to kilometres (1.52) × 150 000 000 (1)	Seen anywhere in the answer Incorrect information shown to be used correctly (1) Correct answer, no working scores full marks 228 000 000 / 2.28×10^8 (km) 228 to any power of 10, allow 1 mark if no other mark awarded. 225 000 000 / 2.25×10^8 (km), allow max 1 mark if no working shown.	**(2)**

Q2.

	Answer	Acceptable answers	Mark

(i)	[redacted] Mercury 4 900 km Venus 12 100 km Mars 6 800 km Three correct (2) One or two correct (1) + or – one square Judge by eye		**(2)**
(ii)	Correct information from table 1.52 (1) Conversion to kilometres (1.52) × 150 000 000 (1)	Seen anywhere in the answer Incorrect information shown to be used correctly (1) Correct answer, no working scores full marks 228 000 000 / 2.28×10^8 (km) 228 to any power of 10, allow 1 mark if no other mark awarded. 225 000 000 / 2.25×10^8 (km), allow max 1 mark if no working shown.	**(2)**

Q3.

		Indicative Content	**Mark**
		A description including some of the following points • improved QUALITY eg higher or better magnification/ detail/resolution or clearer/brighter image **OR** MORE INFORMATION (than with naked	**(6)** **Exp**

		eye) of image/data eg new planets/stars/nebulae/pulsars (This could be extra detail for greater magnification/resolution only) • detection of (non-visible) electromagnetic WAVES eg Xray / UV / IR/ radio • TECHNOLOGY that enable collection of more data eg reflecting telescope/arrays ***and/or*** additions eg computer-aided /photographic connections or larger (objective) lens/mirror • POSITION of telescopes – eg orbital/outside atmosphere/on top of mountains/away from atmosphere/rays not absorbed/obscured/scattered by atmosphere. Ignore 'Hubble' or 'Compton'.	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited description e.g. mention of any one example such as "magnifies stars/planets" OR "discovering new planets/stars" • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple description e.g. mention of either two of the improvements **OR** extra detail about one of the improvements eg improvement **plus** example (ie Magnifies planets *so that craters/mountains may be seen)* • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately	

		• spelling, punctuation and grammar are used with some accuracy
3	**5 - 6**	• a detailed description e.g. mention of three (or more) improvements **OR** two improvements **plus extra detail about one of them (ie Telescopes in space can detect XRays *that would be absorbed by the atmosphere*)** • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors

(Suitable extra detail shown in italics in examples above)

Q4.

	Answer	**Acceptable answers**	**Mark**
(i)	Milky Way	Accept any spelling	**(1)**
(ii)	☒ **D** white dwarf		**(1)**

Q5.

Question Number	**Answer**	**Additional guidance**	**Mark**
(i)	examples: planets have moons (1) the Earth rotates (spins) (1) planets orbit the Sun (1) Pluto is no longer a planet (1) orbits are elliptical (not circular) (1) there are more planets than previously thought (1) ours is not the only solar system (1) Earth is {round/spherical/not flat} (1) planets are not wandering stars.	answers must be to do with the solar system	**(1)**

Question Number:	Answer	Additional guidance	Mark
(ii)	smooth curve drawn on the graph (1) horizontal line drawn from 4.6 Earth years to intercept the drawn line/curve (1) EITHER: their reading from line / curve (1) OR 430±30 (million km) (1)	accept curve up to Mars followed by a straight line plot point at year length = 4.6 reading on distance axis ± half small square from their graph award full marks for the correct answer without working	(3)

Q6.

Question Number	Answer	Additional guidance	Mark
(i)	a description to include: wavelength (of e.m. radiation) increased / frequency decreased (1) as the (star) moves away (from us) (1)	don't penalise planet instead of object stretched/moves to(wards) red end of spectrum spectral lines move towards the red end of the spectrum	(2) AO 1 1

Question Number	Answer	Additional guidance	Mark
(ii)	• an explanation linking: • • big bang has expanding universe (1) • with galaxies moving away (from each other) (1)	 from (original) explosion started at a point accept stars moving away (not objects or planets here) the further away they are the greater is their (recessional) speed idea	(2) AO 1 1

Question Number	Answer	Additional guidance	Mark
(iii)	microwave		(1) AO 1 1

Question Number	Answer	Additional guidance	Mark
(iv)	(radiation) that comes from all over the sky / space / the universe	from the big bang / explosion	(1) AO 1 1

Question Number	Answer	Additional guidance	Mark
(v)	an explanation linking: • the Big Bang theory has a beginning / initial explosion (1) • that releases / gives out radiation (1)	 explosion from a point radiation still present	(2) AO 1 1

Q7.

	Answer	Acceptable answers	Mark
(i)	cosmic microwave background		(1)

	(radiation) all three words needed		
(ii)	(the) Big Bang (theory)		**(1)**
(iii)	change in the (observed) frequency **or** wavelength of light/radiation (received from a distant galaxy)	light/absorption lines is/are shifted toward red end of spectrum light/radiation has longer wavelength/lower frequency waves are more spread out/stretched Ignore references to sound or colour of light eg galaxies/light appear(s) red(er)	**(1)**
(iv)	An explanation linking the following:- It/the Universe is expanding (1) some galaxies are (moving)faster (than others) (1)	they/ galaxies are moving further/away (from the Earth/us) the further away the galaxy is the greater the red-shift/the faster the galaxy is moving (some galaxies) are moving away faster (than others) gains 2 marks IGNORE references to planets/stars	**(2)**

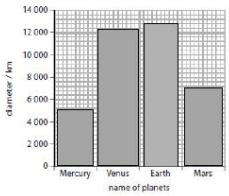
Q8.

	Answer	**Acceptable answers**	**Mark**
(a)(i)	☒D ROS		**(1)**
(a)(ii)	☒C changes changes		**(1)**
(b)(i)	An explanation linking **three** of the following: • the (objective) lens (1) • collects light (from Jupiter) (1) • focuses/converges/refracts the light (1) • (objective) produces a real image (1) • (at the) focal point (1)	Marks can be awarded for a labelled diagram	**(3)**

	• (Image is viewed by the) eyepiece lens (1) • (Eyepiece) produces a virtual image (1)		
(b)(ii)	An explanation linking the following: (the geocentric model) • has the Earth at the centre of the Universe/solar system / everything orbits the Earth (1)(but Galileo observed) • moons of Jupiter (1) • (so) not all things orbit the Earth/ geocentric model is wrong (1)	**Accept alternative argument:** Galileo observed moons/stars/objects Orbiting/going round Jupiter (this) suggested that not everything orbits the Earth	**(3)**
(c)	2100/500 = 4.2 (1) 4.2 × 150 = 630 (million km) (1) Accept ratios as speed is constant 150/500 = distance to Jupiter/2100 OR Distance to Jupiter = (150/500) × 2100 Either for 1 mark	Power of 10 error maximum of 1 mark (speed of light) about 150 000 000 ÷ 500 = 300 000 (km/s) (1) (distance to Jupiter)= 300 000 × 2 100 = 630 000 000 km (1) / = 630 (million km) An answer with no calculation of 630 (million km) gains 2 marks If an answer of 630 million/ 630 000 000 is given with correct working award both marks	**(2)**

Q9.

	Answer	**Acceptable answers**	**Mark**
(a)(i)	Milky Way	Accept any spelling	**(1)**
(a)(ii)	☒ **D** white dwarf		**(1)**

(b)(i)	 Mercury 4 900 km Venus 12 100 km Mars 6 800 km Three correct (2) One or two correct (1) + or – one square Judge by eye		**(2)**
(b)(ii)	Correct information from table 1.52 (1) Conversion to kilometres (1.52) × 150 000 000 (1)	Seen anywhere in the answer Incorrect information shown to be used correctly (1) Correct answer, no working scores full marks 228 000 000 / 2.28×10^8 (km) 228 to any power of 10, allow 1 mark if no other mark awarded. 225 000 000 / 2.25×10^8 (km), allow max 1 mark if no working shown.	**(2)**

		Indicative Content	**Mark**
QWC	***(c)**	A description including some of the following points In Solar System • Use of telescopes • Search for evidence of conditions needed for life e.g. water/oxygen/bacteria on other planets • Manned missions (to the Moon) • Unmanned missions/probes to other planets e.g. Mars, Jupiter, Saturn, Mercury • Landers / robots / rovers sample soil • Information transmitted back to	**(6)**

		Earth Throughout the Universe • Search for extra-terrestrial intelligence (SETI) • Use radio telescopes • Search for (regular pattern of) radio signals • Search for other planetary systems • Discovery of other planetary systems (capable of supporting life) • Broadcasting signals/ sending out messages (to extra terrestrial intelligences)	
Level	**0**	No rewardable content	
1	**1 - 2**	• A limited description giving a (named) way of searching for evidence e.g. SETI **OR** using telescopes **OR** send messages to space **OR** spacecraft • The answer communicates ideas using simple language and uses limited scientific terminology • Spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• A simple description of any TWO of the searches for evidence e.g space probes go to other planets and telescopes are used **OR** radio telescopes and looking for radio waves from space. • The answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • Spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• A detailed description of searches for evidence within **AND** outside the solar system e.g. Unmanned space probes go to other planets and radio telescopes search for radio signals from space. • The answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • Spelling, punctuation and grammar are used with few errors	

Q10.

Question Number	Answer	Acceptable answers	Mark
(c)(i)	Correct plotting (1)	+/- ½ a small square if line is drawn exactly through the point accept for the mark even if point is not obvious	**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)(ii)	Line of best fit drawn	straight line to be within lower two printed dots and upper 3 printed dots does not need to pass through origin **ignore** line below the given points	**(1)**

Question Number	Answer	Acceptable answers	Mark
(d)	A description including		**(2)**
	1. expansion (of space) (1)	**ignore** expansion of Earth, particles and other objects unqualified 'explosion' is insufficient, a reference to expansion is needed	
	and any one of		
	2. continuing (expansion) (1)	(this point only is **dependent** on first)	
	3. from very {hot/dense} start (1)		
	4. from a {point /small volume} (1)	singularity	
	5. origin of Universe (1)	{Universe / Space} still expanding = 2 marks	

Q11.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	B red giant (1)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	C the Milky Way (1)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(iii)	D Proxima Centauri (1)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	description to include: • method (1) • relevant detail (1)	Telescope (inc. radio telescopes) Lander (e.g. robots/drones) Orbiter / Satellite has camera / takes photos / collecting samples (e.g rocks) / analyse atmosphere / climate / signs of water / gases that will support life / can test for water/nutrients ignore repeat of stem	**(2)**

Question Number	Answer	Acceptable answers	Mark
(c)(i)	explanation linking two from: • (on Earth) image is distorted / image not bright enough (ORA) (1) • planets very small / far away (1) • atmosphere (in way) / light pollution (1) • can detect different parts of em spectrum (that are not detectable on Earth) (1) • can keep it pointed at the same spot more easily (1)	Reverse arguments apply throughout (above atmosphere gives) more defined / clearer / better image obscured by clouds waves can be detected (that are not detectable on Earth) not affected by Earth's rotation	**(2)**

Question Number	Answer	Acceptable answers	Mark
(c)(ii)	Suggestion: • planet takes 150 days to orbit the star (1)	has 150 days in a year	**(1)**